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Exploring data augmentation methods to enhance EEG measures for epilepsy seizure detection

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ABSTRACT

Automatic seizure detection using machine learning can reduce the workload of clinicians in epilepsy diagnosis. However, the class imbalance between seizure and non-seizure data limits model performance. Data augmentation offers a solution, yet few studies have systematically compared different augmentation strategies for seizure classification. In this study, we evaluate 12 data augmentation methods across multiple classifiers using EEG data. Beyond accuracy, we assess waveform preservation, spectral consistency, runtime, and feature separability. Results show that Magnitude Warping (MagWarp), Scaling, and Scaling for Multiple Channels (ScalingMulti) consistently yield superior performance. These findings provide practical insights into selecting effective augmentation techniques for real-world epilepsy detection systems and can support more reliable, automated diagnostic tools in clinical practice.

1. Introduction

Epilepsy is a neurological disorder characterized by recurrent seizures caused by the overexcitation of neurons in the brain [1]. It is one of the most common neurological disorders, affecting over 50 million people worldwide [2]. The severity and impact of epileptic seizures can vary greatly, ranging from brief lapses in attention to severe convulsions. Due to the unpredictability and potential severity of seizures, accurate diagnosis and monitoring are critical for effective management and treatment of epilepsy. In clinical applications, electroencephalogram (EEG) is the most common approach to epileptic seizure detecting and epileptic lesion locating. It is a non-invasive recording of electrical activity in the brain that can reveal abnormal discharges indicative of seizures. However, the interpretation of epileptic events usually requires specialized clinicians to examine several tens of hours of EEG data, which is extremely time-consuming and experience-dependent. Therefore, the development of highly sensitive automatic seizure detection algorithms is essential to assist clinicians, reducing their workload and improving diagnostic accuracy.

Epileptic seizure detection is a distinctly unbalanced binary classification task. In common epilepsy seizure datasets, the amount of normal EEG data highly exceeds that of seizure episodes. For instance, in the most commonly used CHB-MIT database, the proportion of seizure episodes is extremely low, with the ratio of seizure to non-seizure periods being 1 to 931 [3]. The significant class imbalance causes issues for model training, as they are biased towards the majority (non-seizure) class, thereby reducing the sensitivity (SEN) of seizure detection. More seriously, in the clinical application of epilepsy detection, sensitivity is far more important than specificity, or even than overall accuracy.

The class imbalance issue is vital for effective seizure detection, and can be potentially solved at model or data level [4]. At the model level, many advanced algorithms have been proposed, such as adjusting class weights during model training. These methods have been proven effective in many studies [5,6]. By contrast, solving class imbalance issue through data augmentation at the data level is simpler and more universal. It can be compatible with any model improvement method. Data augmentation aims to balance the dataset by increasing the number of minority class samples, enhancing feature representation and increasing the diversity of training samples.

Several data augmentation techniques, including Jittering [7], Scaling [8], and Permutation [9], have been applied to boost the performance of EEG-based seizure detection models. Although a wide variety of data augmentation methods have been proposed for this task, prior studies often focus on a small subset of techniques, adopt inconsistent evaluation protocols, and validate performance on a single classifier or dataset. In most cases, the assessment is limited to classification accuracy, offering limited insight into how augmentation influences signal structure or feature representation.

To address these limitations, our study provides a comprehensive evaluation of 12 data augmentation methods under a unified

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framework across multiple classifiers and datasets. Our analysis incorporates classification performance, waveform characteristics, spectral distortion, runtime, and feature separability. The results indicate that Magnitude Warping (MagWarp), Scaling, and Scaling for Multiple Channels (ScalingMulti) are consistently the most effective augmentation strategies for seizure detection. Our study offers practical guidance for optimizing data augmentation in EEG-based clinical applications.

2. Data augmentation methods for EEG data

In this study, we employed 12 different data augmentation techniques to address the class imbalance issue in epilepsy seizure detection. These methods include conventional strategies such as Jittering, Scaling, and Permutation, as well as more advanced techniques, to enhance feature representation and increase the diversity of training samples. By systematically evaluating these approaches, we aimed to identify the most effective data augmentation method for improving the performance of epilepsy seizure detection models.

(1) **No extra procedures with simple duplication (NO)**: This method directly duplicates the original minority class data, which specifically refers to epilepsy seizure data. Its primary objective is to ensure that the quantity of training data aligns with that of data processed using other data augmentation strategies, thereby facilitating subsequent performance comparisons.

(2) **Jittering**: This method introduces Gaussian noise or slight offsets to expand the minority class data in the dataset [10].

(3) **Scaling**: This technique increases the amount of minority class data by applying a random scalar to the original signal [11].

(4) **Scaling for Multiple Channels (ScalingMulti)**: This approach achieves data augmentation by multiplying the original signals across different channels using the same random scalar [11].

(5) **Permutation**: This approach achieves data augmentation by randomly segmenting the signal and then recombining the segments in a random order [11].

(6) **MagWarp**: This method achieves data augmentation by convolving the original data with a curve of random values set to 1 [12].

(7) **Time Warping (TimeWarp)**: This method achieves data augmentation by applying nonlinear transformations to the time axis, thereby increasing the diversity of the dataset. Specifically, it changes the positions of data points in the time series, causing the sequence to warp in time while preserving its shape characteristics [13]. The generated random warps consist of Gaussian noise with a mean value of 1 and a standard deviation of 0.2. The number of knots for spline interpolation is 3.

(8) **Window Slicing (WinSlice)**: This method achieves data augmentation by slicing samples and then resizing them to their original length using interpolation [14].

(9) **Window Warping (WinWarp)**: This method applies a 10% distortion to time windows to enhance data variability by warping or compressing the time axis within selected windows [13].

(10) **Window Warping for Multiple Channels (WinWarpMulti)**: This method applies a 10% distortion to time windows to enhance data variability by warping or compressing the time axis within selected windows across different channels using the same random scalar [13].

(11) **Random Guided Warping (RGW)**: This method is a data augmentation method based on the Dynamic Time Warping (DTW) algorithm [15]. DTW is used to compare the similarity between two time series, handling series of different lengths and allowing nonlinear time alignment. The core concept involves aligning corresponding elements of two sequences along the time axis to find the alignment path that minimizes the distance between them.

The steps of the algorithm are as follows: (a) Construct a distance matrix to store the distances between each pair of elements in the two time series. (b) Initialize the first row and column of the matrix so that the cumulative distance progressively increases along these axes. (c) Gradually compute the minimum cumulative distance for each cell to fill the entire distance matrix. (d) Trace back through the matrix to find the optimal alignment path based on the minimum cumulative distances. (e) Determine the minimum distance between the two time series based on the optimal alignment path.

The specific implementation of RGW is as follows: To generate an augmented signal that maintains the characteristic distribution of the original signal, RGW utilizes guidance from other reference patterns in the time domain to appropriately warp the signal. Assuming the training dataset S contains two different data elements, s and r, a new signal s' is generated using both sequences. s' retains the characteristics of the student time series s while adopting the step size of the teacher time series r. The operation is executed as follows:

The minimum cumulative sum matrix for DTW is calculated using Eq. (1):

$$D(i,j) = C(r_i, s_j) + \min_{(i',j') \in \{(i,j-1),(i-1,j),(i-1,j-1)\}} D(i',j') \tag{1}$$

where $D(i,j)$ is the cumulative sum of the ith and jth elements, and $C(r_i, s_j)$ is the local distance between r_i and s_j. By backtracking from $D(I,J)$ to the origin $D(0,0)$ through the matched elements (i',j'), the minimum alignment path is identified. Next, the time series s is warped according to time steps $1, \ldots, j^i, \ldots, J^i$ to create a new time series s'. This new sequence s' possesses the characteristic values of s but adopts the time step of r under the constraints of the DTW-guided warping path. Using this method, N^2 time series can be synthesized, where N is the number of samples during the epileptic seizure period.

(12) **RGW with shapeDTW (RGWs)**: The primary distinction of this method lies in its use of shapeDTW rather than the traditional DTW algorithm [15]. Unlike DTW, which matches time series elements on a point-by-point basis, shapeDTW enhances alignment by dynamically matching shape descriptors within the sequences [16]. To elaborate, for a given time series s, sub-sequences of length W are sampled. These sub-sequences are denoted by $d_{s_j} = (s_{j-\lceil \frac{1}{2}W \rceil}, \ldots, s_j, \ldots, s_{j+\lfloor \frac{1}{2}W \rfloor})^T$. With a step size of 1, this process results in a new sequence of shape descriptors $d_s = d_{s_1}, \ldots, d_{s_j}, \ldots, d_{s_J}$, maintaining the same length as the original series s.

The computational process for shapeDTW mirrors that of standard DTW but shifts focus from calculating distances between original individual elements to those between the sequences of shape descriptors d_s and d_p. This adjustment means that the time series s and r are now aligned using their shape descriptors, and the cost function is defined by the distances between these descriptors, given by $C_{r_i,s_j} = d_{r_i} - d_{s_j}$.

Apart from the introduction of shapeDTW, the computational procedure for RGWs remains identical to that of RGW and is not detailed further.

Fig. 1 visualizes the augmented data generated by 12 different data augmentation methods.

3. Experiments

3.1. Dataset

The CHB-MIT database, which was publicly released in 2010 by researchers from the Massachusetts Institute of Technology, contains continuous EEG data collected over several hours to days from epileptic patients at Boston Children's Hospital [17]. The database includes 24 records from 23 patients, including two separate records (chb01 and chb21) from the same female patient, collected 1.5 years apart. All data in this collection were acquired using the 10–20 international system for electrode placements. In our study, we used 18 channels consistently across all 23 patients. These channels are most commonly used for epilepsy research, including FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, and CZ-PZ [18].

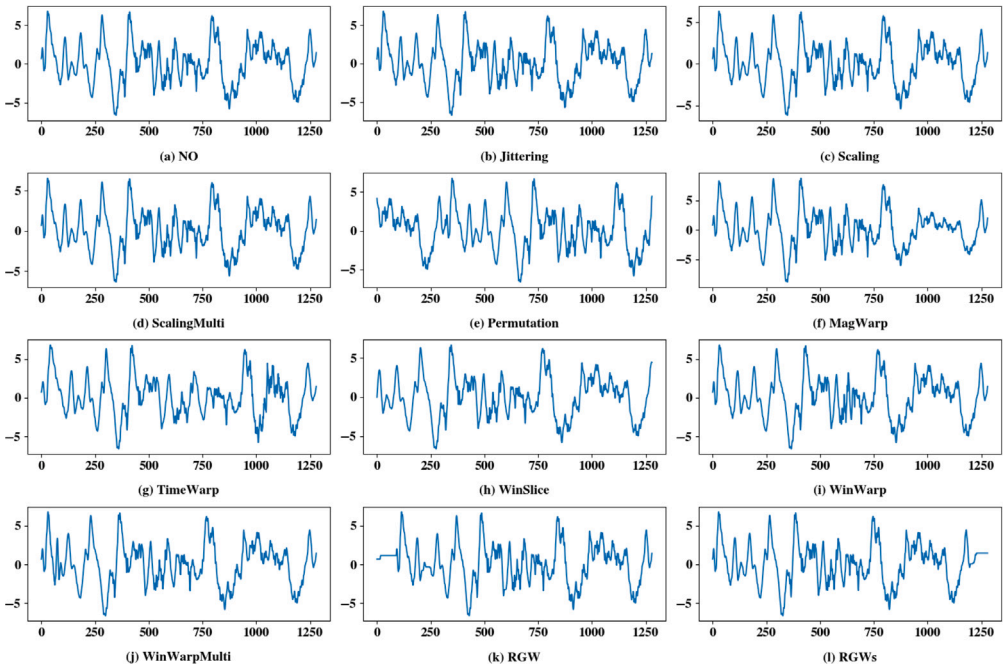


Fig. 1. Augmented Data Generated by 12 Data Augmentation Methods. This visualization highlights the differences in temporal continuity and waveform morphology introduced by different augmentation strategies, which may impact model performance. Notably, methods such as Jittering and Permutation produce more distorted waveforms compared to others like Scaling or ScalingMulti.

Table 1
The results on the CHB-MIT dataset, including SEN, ACC, and AUC, were obtained using 12 data augmentation methods on 5 classifiers at a 100x augmentation ratio.

Method	KNN			LRC			RFC			DTC			AdaBoost		
	SEN	ACC	AUC	SEN	ACC	AUC	SEN	ACC	AUC	SEN	ACC	AUC	SEN	ACC	AUC
NO	70.87	**97.90**	84.67	79.53	96.38	88.12	60.94	98.91	80.35	56.79	**98.55**	78.07	71.66	98.17	85.22
Jittering	69.08	**97.70**	83.69	73.10	**97.58**	85.58	63.79	98.85	81.71	58.44	97.59	78.41	71.86	**98.28**	85.36
Scaling	**77.97**	96.40	**87.36**	**83.38**	95.93	**89.80**	**81.69**	98.50	**90.29**	**79.08**	95.86	**87.65**	**85.93**	96.49	**91.35**
ScalingMulti	75.11	97.39	86.49	79.87	96.42	88.31	69.23	**98.94**	84.42	65.37	97.9	81.97	79.81	97.91	89.10
Permutation	72.67	97.57	85.37	80.48	96.16	88.47	66.09	98.93	82.88	59.38	**98.52**	79.35	76.15	**98.18**	87.41
MagWarp	**77.67**	96.43	**87.23**	**83.51**	95.93	**89.86**	**81.50**	98.51	**90.21**	**78.76**	96.55	**87.84**	**84.43**	97.17	**90.95**
TimeWarp	73.96	96.62	85.51	74.60	96.97	86.01	80.76	96.79	88.96	75.20	92.86	84.18	83.84	95.70	89.85
WinSlice	73.88	97.39	85.89	75.96	**96.98**	86.64	67.04	**98.94**	83.33	66.30	97.61	82.27	79.57	97.12	88.55
WinWarp	73.96	97.33	85.89	81.20	96.31	88.90	70.03	98.94	84.82	65.8	98.18	82.32	79.74	97.76	88.97
WinWarpMulti	73.97	97.35	85.90	81.13	96.27	88.85	69.96	98.94	84.78	64.37	98.06	81.56	76.81	97.93	87.59
RGW	73.52	97.45	85.73	78.75	96.28	87.69	67.02	98.91	83.33	61.61	98.28	80.32	77.19	98.04	87.86
RGWs	73.05	97.51	85.53	80.36	96.36	88.52	67.60	98.92	83.62	60.89	**98.34**	80.00	76.25	97.90	87.32

3.2. Experimental procedures

The experiment consisted of four steps: data preprocessing, data augmentation, feature extraction, and classification. The experimental procedures were illustrated as follows:

(1) This work adopted a leave-one-out cross-validation strategy. Since N records were recorded for epilepsy patients (one record refers to one EDF file in the CHB-MIT database), the model performance was validated over N rounds of testing [19,20]. In each round, the data were divided into two groups: one group comprised training segments obtained from $N-1$ records, and the other group comprised testing segments from the last EEG record. (2) The 12 data augmentation techniques mentioned in the last section were performed on the training data. According to the clinical applications, the testing data were not augmented. (3) Feature extraction was performed separately on the training and testing datasets. Initially, Discrete Wavelet Transform (DWT) was applied to decompose the original signal into four sub-bands. DWT is a mathematical tool used to decompose a signal into its frequency sub-bands, retaining both time and frequency information. To select the most informative features, we conducted a preliminary feature selection experiment on a subset of data. The following 11 features were extracted across 18 channels and 4 sub-bands: variance, energy, Teager–Kaiser energy operator, maximum, mean, skewness, kurtosis, line length, Shannon entropy, sample entropy, and Hurst exponent [21]. Based on their overall ranking, the top three features, signal energy, variance, and Teager–Kaiser energy operator, were selected. As a result, each sample was represented by a 216-dimensional feature vector (18 channels × 4 sub-bands × 3 features) was extracted for each sample. (4) In our study, five classifiers, including k-nearest Neighbors (KNN) [22], Linear Regression Classifier (LRC) [23], Random Forest Classifier (RFC) [22], Decision Tree Classifier (DTC) [23], and Adaptive Boosting (AdaBoost) [24], were applied. These classifiers were implemented using the sklearn library, with all parameters set to their default values. (5) Finally, this study used three evaluation metrics—Accuracy (ACC), Sensitivity (SEN) and the area under the Receiver Operating Characteristic curve (AUC), to compare the 12 data augmentation methods. Given the strong class imbalance in the epilepsy seizure dataset, both SEN and AUC provide more informative and reliable assessments than overall accuracy. SEN reflects the model's ability to correctly identify seizure events, which is particularly critical in clinical applications where false negatives may lead to delayed intervention or increased patient risk. AUC offers a threshold-independent evaluation of the model's discriminative capacity and is robust to class imbalance. While specificity measures the correct identification of non-seizure

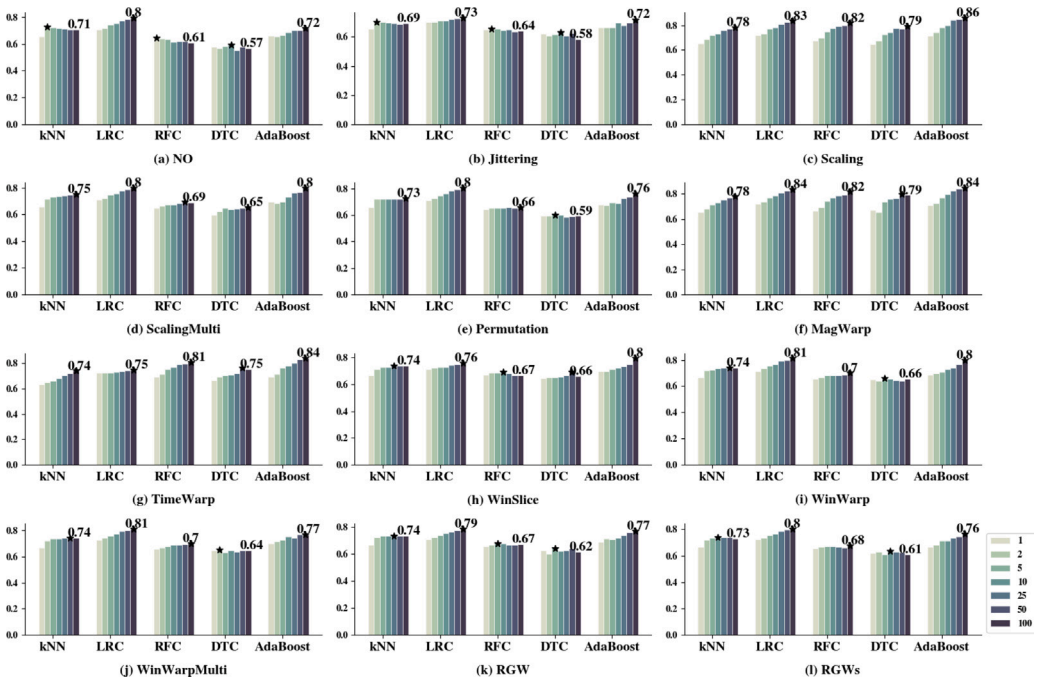


Fig. 2. The SEN were obtained using 12 data augmentation methods on 5 classifiers at 7 different augmentation ratio.

segments, it is less emphasized in this context because false positives are generally more tolerable than missed seizures. Therefore, in our subsequent analysis, we primarily focus on SEN and AUC as the most clinically meaningful performance indicators.

To facilitate reproducibility and future research, the implementation code and relevant resources have been publicly released at https://github.com/Open-EMG/EEG-AUGBENCH.

4. Results

4.1. The results at 100x augmentation ratio

Table 1 presents the average results obtained across 12 data augmentation techniques and five classifiers, using a default configuration of 100× augmentation. This setting is selected as it offers a practical trade-off between improving classification performance and maintaining computational efficiency. To highlight the best-performing algorithms, the top three data augmentation methods for each classifier are marked as follows: bold and underline, bold, and underline, respectively. From Table 1, five findings can be observed: (1) Scaling, MagWarp, and TimeWarp perform the best. Scaling and MargWarp rank in the top two for both SEN and AUC. Applying TimeWarp, coupled with the RFC, DTC, and AdaBoost, results in the third-highest rankings for SEN and AUC. (2) Although Scaling and ScalingMulti are similar data augmentation methods, Scaling proves to be more effective than ScalingMulti. ScalingMulti multiplies a uniform random scalar across all channels of the original signal, whereas Scaling applies a different random scalar to each channel. Accordingly, scaling increases the diversity of training data. (3) WinWarp and WinWarpMulti differ in whether they uniformly accelerate or decelerate the same location data across different channels. The results show no significant performance difference between these two methods. (4) RGW and RGWs are similar data augmentation methods, differing in the DTW method used. RGW employs the classic DTW, whereas RGWs uses the improved shapeDTW, which has higher time complexity. However, no significant performance difference is observed after augmentation. (5) Jittering and Permutation, the two most conventional methods, show poor performance in terms of SEN and AUC.

4.2. Classifier performance across different augmentation techniques

To clearly demonstrate the performance of various data augmentation methods across different classifiers, Fig. 2 depicts the SEN results of 12 data augmentation methods using five classifiers. The specific SEN at augmentation ratios of 100 are highlighted in bold numbers, whereas asterisks indicate the highest SEN achieved by a classifier at a specific augmentation ratio of k. For instance, Fig. 2(a) shows that using the NO method combined with the KNN classifier at an augmentation ratio of 2 results in better SEN than using other augmentation ratios including 1, 5, 10, 25, 50, and 100.

Based on Fig. 2, we have the following findings: (1) Regardless of the classifier used, it is observed that when the data augmentation ratio is increased to 50 or 100, the performance of Magwarp, Scaling, and ScalingMulti is the most outstanding. This proves that using these methods to continue generating more minority class data still has room for performance improvement. Conversely, the original NO method simply replicates the original minority class data, resulting in lower feature set diversity. This limited diversity means that generating more minority class data leads to only marginal improvements or even a decline in performance. For instance, when using KNN as the classifier, increasing the data augmentation to 100 times only slightly outperforms a 1-fold augmentation, and is significantly worse than a 2-fold augmentation. (2) When using effective data augmentation methods such as Magwarp, Scaling and ScalingMulti, AdaBoost achieves the best sensitivity, surpassing the LRC algorithm. This indicates that AdaBoost can achieve superior classification performance when data augmentation is highly effective. (3) Conversely, when using less effective data augmentation methods such as NO, Jitter, Permutation, RGW, and RGWs, it is found that the LRC achieves the best sensitivity, surpassing the AdaBoost algorithm. This suggests that LRC can achieve superior classification performance when the data is imbalanced or when there is a lack of diversity in the minority class data.

4.3. Comparative analysis of augmentation techniques across diverse rates

Fig. 3 depicts the average ranking for 12 methods under different data augmentation ratios. From Fig. 3, we can discern the following trends: (1) As the data augmentation ratio increases, the AUC rankings

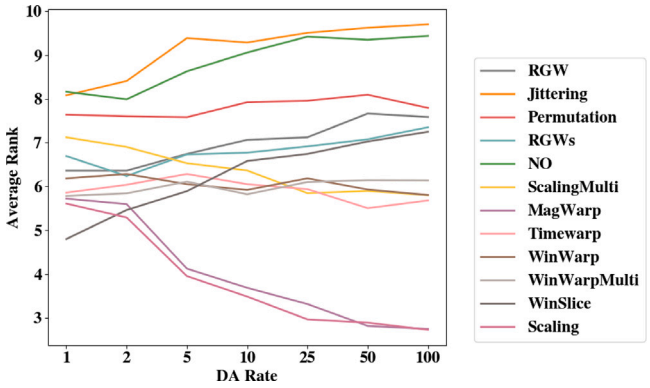


Fig. 3. The average rankings of the 12 methods change as the data augmentation ratio increases.

for the Scaling and MagWarp methods progressively improve. (2) With increasing enhancement ratios, the AUC rankings for the NO and Jittering methods gradually decline. (3) In most DA ratios, the performance of Scaling is superior to that of ScalingMulti, and RGWs are superior to RGW.

4.4. Comparing augmented data and their generated PSD plots

Fig. 4 visualizes the Power Spectral Density (PSD) of DA data obtained through 12 different data augmentation methods. The implementation is based on Welch's method, which involves segmenting the signal into multiple overlapping sections, computing the Fourier transform for each segment, and then averaging the power spectra of these segments to obtain a smoothed PSD estimate. From the figure, it can be observed that the PSDs of Scaling, ScalingMulti, and MagWarp methods closely resemble the overall trend and local details of the PSD of the original signal.

4.5. Comparing separability of features extracted by different DA methods

To further investigate the separability of features extracted by different data augmentation methods, the variance of all features was calculated as shown in Fig. 5. In the field of data augmentation, we aim for augmented data to exhibit high feature separability. A straightforward method to assess feature separability is by calculating the variance of features, with a higher average feature variance indicating greater separability across the dataset. The specific calculation method is as follows: the variance of each feature for each patient is calculated, and then the average is computed. For easier visualization, the computed average variance is then normalized. The feature variance visualization reveals that MagWarp, TimeWarp, and Scaling have relatively high dispersion, corresponding with their superior experimental results. Additionally, we observe that ScalingMulti shows lower separability compared to Scaling, indicating that ScalingMulti has reduced ability to differentiate between different categories or features. Similarly, WinWarpMulti exhibits lower separability than WinWarp, suggesting that the introduction of the "Multi" version has weakened the model's capability to distinguish between categories. Furthermore, we notice that RGW has lower separability compared to RGWs, implying that the RGW model performs worse than RGWs in distinguishing specific data categories. This consistency is also reflected in the accuracy results, where the former of each similar method pair tends to perform worse. Thus, we can conclude that data variance after augmentation is an important indicator to measure the effectiveness of the data augmentation method.

Furthermore, we found that the Gini index can also measure the distribution uniformity and dispersion of feature values. For instance, if samples have similar values, the Gini coefficient is small, indicating no dispersion. If the data have diverse distribution, the Gini coefficient becomes higher, indicating high dispersion. Similarly, for ease of visualization, the computed average Gini index is also normalized, as shown in Fig. 5. The Gini index results show that the features generated by the NO, Permutation, RGW, and RGWs methods have the lowest Gini index, corresponding to their poor performance across the 12 methods. The Gini indices of the other 8 methods are similar, making it difficult to distinguish their feature impurity. Therefore, the Gini index can effectively identify methods with lower feature separability.

Table 2
Runtime (mean ± std) of different augmentation methods.

Method	Runtime (ms)
Jittering	0.491 ± 0.059
Scaling	0.021 ± 0.006
Permutation	0.116 ± 0.044
MagWarp	2.875 ± 0.268
TimeWarp	3.415 ± 0.039
WindowSlice	0.444 ± 0.012
WinWarp	0.748 ± 0.013
ScalingMulti	0.027 ± 0.016
WinWarpMulti	0.907 ± 0.062
RGW	230 ± 1.4
RGWs	1830 ± 35.5

4.6. Comparison of running times for different methods

The running times for applying 11 different data augmentation methods to two samplings (since RGW and RGWs require at least two samplings to perform data augmentation) are shown in Table 2. The Scaling and ScalingMulti methods, which only require scalar multiplication, as well as the Permutation method, which involves random segmentation and recombination, have the shortest running times. The RGW and RGWs methods, which require the calculation of DTW and shapeDTW distances, have the longest running times.

4.7. Supplementary evaluation on SWEC-ETHZ

To further evaluate the generalizability of the proposed augmentation methods, we conducted an additional set of experiments on the SWEC-ETHZ dataset [25]. The experimental details and results are provided in Appendix.

5. Discussion

Overall, the MagWarp, Scaling, and ScalingMulti methods exhibit the best performance. In addition to their visibly superior performance, all other results also demonstrate strong advantages. First, the augmented data generated by these three methods (as shown in Fig. 1(f), (c), and (d)) are visually similar to the original data. Second, PSD of the augmented data generated by these methods also shows high similarity to the original data. For example, in Fig. 4(a), the original data shows a local peak around 60 Hz, which may be due to insufficient power line filtering. These three methods (Fig. 4(f), (c), and (d)) retain this local frequency detail. Many previous studies have also reported the strong data augmentation capabilities of these three methods. Comparing these methods, MagWarp is the most recommended without considering time complexity. However, once time complexity is considered, the Scaling method, which only requires multiplying each sample point by a fixed scalar, becomes more favorable due to its simplicity. The ScalingMulti method, which multiplies different channels by fixed scalars, reduces feature separation (see Fig. 5), thus decreasing classification performance.

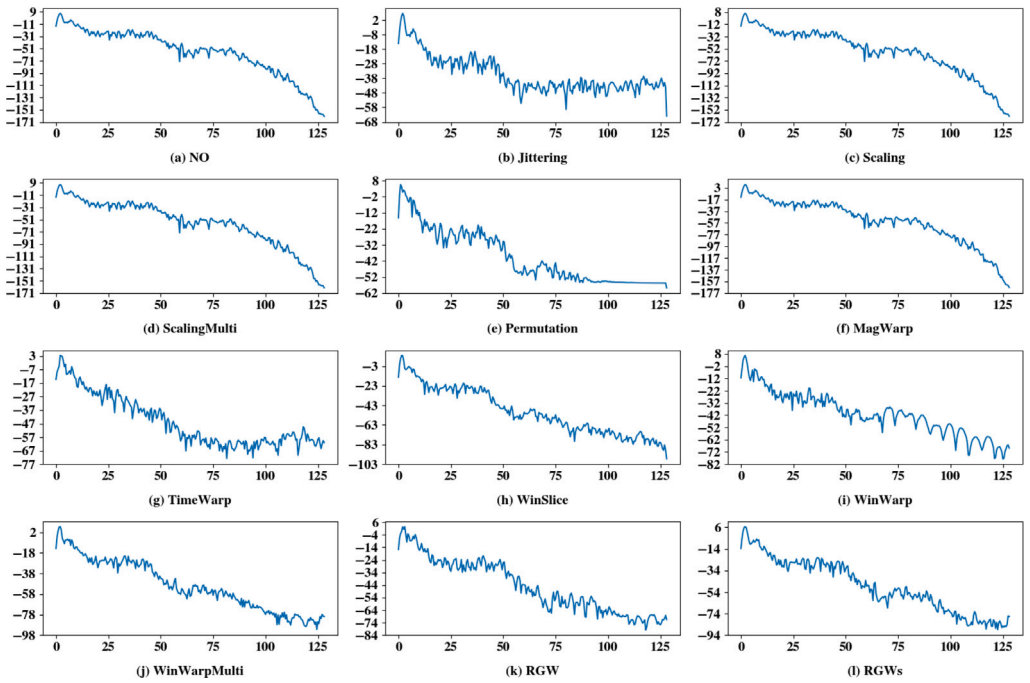


Fig. 4. PSD Plots Generated from Augmented Data Using 12 Data Augmentation Methods.

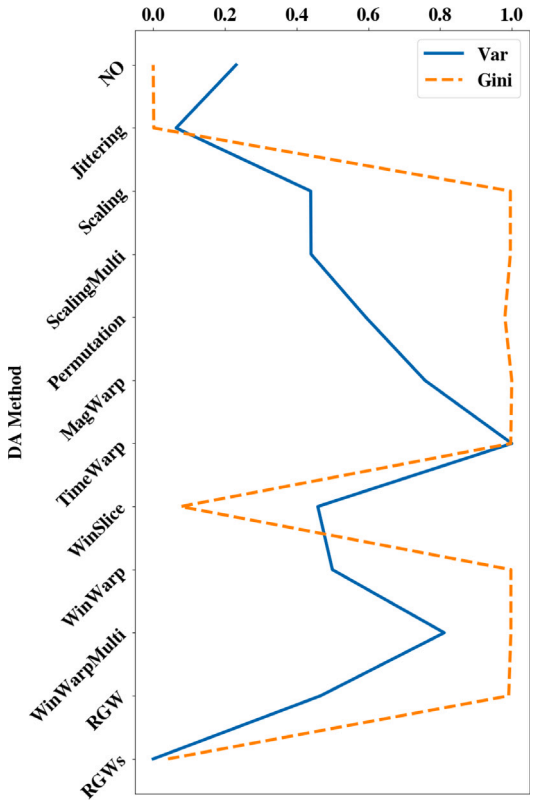


Fig. 5. Normalized variance (Var) and Gini index (Gini) of all features obtained using 12 data augmentation methods. Both metrics are normalized to [0, 1] for fair comparison across methods.

Among the 12 evaluated data augmentation methods, Jittering, Permutation, RGW, and RGWs demonstrated notably lower performance in terms of SEN and AUC. Detailed analysis suggests that these methods introduce distortions that compromise the temporal and spectral integrity of EEG signals, which is essential for reliable seizure detection. EEG signals associated with epileptic seizures typically exhibit a characteristic temporal progression. This includes a sharp onset marked by rapidly increasing amplitude and frequency, a sustained phase of high-amplitude activity during the seizure, and a gradual reduction in signal intensity as the episode resolves. This three-phase structure conveys important information in both the time and frequency domains, which is critical for effective feature extraction and classification. In this study, the extracted features primarily include dynamic time-domain descriptors, such as TK, energy, and variance, computed across multiple frequency bands. These features are designed to capture nonstationary changes in amplitude and frequency that reflect the physiological characteristics of seizure events. Jittering introduces small random perturbations at each sampling point of the signal. Although the resulting waveform appears similar in the time domain, the PSD is significantly distorted. As illustrated in Fig. 4, the jittered signal exhibits substantial deviation in spectral content compared to the original, particularly in seizure-related frequency components. This added variability increases intra-class variance and weakens the discriminative power of features like TK, which are highly sensitive to local signal dynamics. Permutation modifies the signal by segmenting and reordering its parts, preserving the amplitude distribution but disrupting the temporal sequence. Since TK energy depends on the continuity of amplitude and frequency transitions between adjacent time points, this temporal distortion considerably reduces its representational capacity. Other temporal features, including entropy and autocorrelation, are also negatively affected. PSD analysis further indicates that permutation flattens spectral variation in the high-frequency range between 80 and 125 Hz, eliminating patterns that are physiologically meaningful for seizure detection. RGW and RGWs are more complex augmentation methods based on dynamic time warping (DTW and shapeDTW). These methods generate new signals by aligning and combining temporal patterns from a pair of original sequences. Although the resulting waveforms are self-consistent, they diverge significantly from the original signals in both time and frequency domains. Moreover, their reliance on student–teacher pairs restricts the total number of unique augmented samples to $N \times (N-1)$. In datasets with limited seizure data, this often results in duplicated samples and a reduction in overall data diversity. The requirement to compute DTW or shapeDTW distances

also incurs substantial computational cost, making these the most time-intensive methods in our evaluation. Despite their lower performance in our benchmark, these augmentation techniques may still be useful in specific circumstances. In low-data scenarios where the risk of overfitting is high, Jittering and Permutation can serve as regularization strategies to introduce variation. When the model or features rely more on global statistical characteristics such as mean or variance rather than temporal structure, the disruptive effects of these augmentations may be reduced. Similarly, RGW and RGWs may provide benefit when the dataset is sufficiently large and balanced, and when the computational resources are adequate to support the generation of synthetic samples that enhance classification near decision boundaries.

We selected five basic classifiers to handle epilepsy detection data, including KNN, LRC, RFC, DTC, and AdaBoost. The experimental results clearly demonstrated significant performance differences across these classifiers in processing the data. Specifically, KNN and LRC showed different levels of effectiveness, especially when dealing with complex boundary data. Epilepsy data, characterized by continuous time series, often has complex and unclear boundary distributions between seizure and normal data. This complexity likely explains why KNN underperformed in this scenario, as it is generally not well-suited for handling data with complex boundaries. RFC, being an improved version of DTC, performed significantly better than DTC in processing the epilepsy data. However, its performance was still not on par with LRC and AdaBoost. This could be attributed to the use of the default setting of 100 trees in RFC for this study. Theoretically, increasing the number of trees could enhance its performance. Nevertheless, as the primary focus of this study was not on optimizing classifier performance but rather on evaluating the overall performance of these data augmentation methods in the context of epilepsy detection, we did not adjust the default settings for RFC. Overall, LRC and AdaBoost both performed exceptionally well in this task. Notably, in scenarios where data augmentation methods, such as Jitter, Permutation, had limited effectiveness, LRC outperformed AdaBoost. This finding suggests that, although AdaBoost is a powerful ensemble method, LRC may be a better choice when data augmentation results are suboptimal. Specifically, when the augmented data does not closely resemble actual seizure data and instead introduces more noise, AdaBoost might overfit these noisy samples. In contrast, LRC, with its regularization capabilities, can better manage these situations, resulting in superior performance.

In this study, we observed that scaling and magnitude warping led to performance improvements in seizure detection, whereas jittering and permutation resulted in performance degradation. Similarly, Li et al. compared jittering, scaling, and permutation on EEG-based fatigue estimation and reported that scaling achieved better performance than the other two methods [8]. Park et al. also compared multiple data augmentation strategies for user-independent fatigue detection using EEG signals and found that while scaling and magnitude warping offered limited gains, jittering consistently harmed performance. [26]. In contrast, Um et al. investigated data augmentation strategies for wearable-sensor-based Parkinson's disease monitoring and observed an opposite trend: permutation yielded the best performance, followed by magnitude warping and scaling, while jittering remained the least effective [27]. This discrepancy can be attributed to the inherent differences in task characteristics. In seizure detection and fatigue estimation, critical pathological events are typically transient and temporally localized, making the preservation of temporal structure essential. As a result, augmentations that disrupt the original sequence order, like permutation, can obscure key temporal features and harm classification accuracy. In contrast, the motor states in Parkinson's disease, such as bradykinesia and dyskinesia, typically exhibit prolonged and stationary signal patterns over extended durations. These patterns are less sensitive to precise temporal alignment within a window, allowing models to benefit from permutation-based augmentation that enhances generalization without compromising class-discriminative features.

This study has several limitations as follows: (1) The results of deep learning-based data augmentation methods or deep neural network classifiers were not presented in this study, primarily due to the limited availability of seizure data per patient. For instance, Subject 2 in the CHB-MIT dataset has only a few seizure events, which results in an insufficient number of positive samples for training deep models effectively. This limitation affects both the reliability of deep generative augmentation techniques and the performance of deep learning classifiers, which typically require large datasets to generalize well. In addition, deep learning approaches are computationally intensive and often require lengthy training and inference times, making them less suitable for real-time or resource-constrained clinical settings. In future work, we plan to explore these approaches in larger datasets with more abundant seizure events to assess their potential under data-rich conditions. (2) Transfer learning and self-supervised learning represent promising strategies for addressing the data scarcity problem inherent in seizure detection tasks. Transfer learning enables models to benefit from knowledge gained in related domains, reducing the dependence on large labeled datasets. Self-supervised learning, on the other hand, can extract meaningful representations from unlabeled EEG data, which is particularly valuable when annotated seizure events are limited. These approaches have the potential to improve model generalization and robustness. In future work, we plan to explore both strategies to enhance the effectiveness of data-driven methods under low-resource clinical conditions. (3) Due to the constraints of the dataset used in this study, which lacks detailed annotations of seizure types, we were unable to investigate the suitability of different data augmentation methods for various types of seizures. Future research could utilize real-world data from hospitals with detailed annotations of seizures and seizure types. This would allow for a more comprehensive analysis of the combined impact of data augmentation methods and seizure types on model performance, ultimately providing more precise and targeted approaches for data augmentation methods.

6. Conclusion

Our primary objective is to investigate the impact of data augmentation algorithms on the performance of epilepsy monitoring models. To achieve this, we employed 12 different data augmentation techniques, encompassing a wide range of data generation methods, from simple transformations, adjustments, and interpolations to the synthesis of new time series data using DWT. Additionally, we applied five different types of classifiers to comprehensively explore the performance of data augmentation techniques across various classifiers. The experimental results indicate that Magwarp, Scaling, and ScalingMulti methods consistently outperform others from multiple perspectives. Furthermore, we found that when effective data augmentation methods are applied, the AdaBoost algorithm achieves the best performance. Conversely, when less effective data augmentation methods are used, the LRC algorithm provides superior classification performance. This study offers researchers a broader selection of data augmentation methods, moving beyond the traditional Jittering approach.

CRediT authorship contribution statement

Yao Guo: Writing – original draft. **Xiaoxiao Zhang:** Writing – review & editing. **Chenyun Dai:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Additional evaluation on the SWEC-ETHZ dataset

Table A.1
The results on the SWEC-ETHZ dataset, including SEN, ACC, and AUC, were obtained using 12 data augmentation methods on 5 classifiers at a 100x augmentation ratio.

Method	KNN			LRC			RFC			DTC			AdaBoost		
	SEN	ACC	AUC	SEN	ACC	AUC	SEN	ACC	AUC	SEN	ACC	AUC	SEN	ACC	AUC
NO	78.40	98.54	88.68	**88.26**	97.43	92.95	65.87	98.67	82.63	70.64	**98.25**	84.75	81.14	98.47	89.98
Jittering	64.60	**99.18**	82.26	76.79	**99.17**	88.25	75.18	**99.03**	87.35	78.92	**98.58**	88.97	77.72	**98.80**	88.50
Scaling	78.73	98.34	88.72	**87.78**	98.53	**93.29**	**83.08**	98.12	**90.73**	**83.20**	94.84	**89.10**	**88.32**	96.85	**92.68**
ScalingMulti	**82.91**	98.37	**90.80**	87.41	98.55	93.13	73.95	98.78	86.61	75.92	97.20	86.76	85.17	98.00	91.72
Permutation	78.43	98.67	88.76	87.19	98.58	93.04	69.85	98.80	84.61	70.81	97.59	84.48	80.67	**98.60**	89.82
MagWarp	78.61	98.42	88.70	87.73	98.53	**93.26**	**83.50**	97.94	**90.86**	**84.98**	95.49	**90.32**	**87.36**	96.98	**92.27**
TimeWarp	72.71	**98.89**	86.05	80.87	**98.99**	90.17	80.72	97.27	89.14	79.74	94.89	87.43	84.41	97.47	91.08
WinSlice	79.59	98.69	89.32	84.28	98.74	91.72	70.92	98.75	85.09	75.34	97.45	86.59	81.25	97.93	89.78
WinWarp	**80.72**	98.37	**89.73**	87.29	98.39	93.00	75.17	98.82	87.24	72.00	97.49	85.00	80.11	98.32	89.41
WinWarpMulti	80.58	98.40	89.68	87.33	98.43	93.04	75.50	98.90	87.45	73.78	97.32	85.80	83.00	98.24	90.79
RGW	79.72	98.52	89.31	86.59	98.52	92.71	69.08	98.82	84.27	73.23	97.66	85.74	79.35	98.36	89.04
RGWs	79.53	98.54	89.23	86.88	98.51	92.85	69.79	**98.94**	84.66	71.39	97.80	84.85	81.51	98.51	90.18

A.1. Dataset description

To further assess the generalizability and robustness of our data augmentation strategies, we performed additional experiments on the SWEC-ETHZ iEEG dataset. This dataset was developed through a collaboration between ETH Zurich and the Sleep-Wake-Epilepsy Center (SWEC) at the University Hospital of Bern. It consists of 2,664 h of continuous intracranial EEG (iEEG) recordings collected from 18 patients with drug-resistant epilepsy and includes 116 clinically annotated seizure events. All seizure onsets and offsets were visually marked by experienced neurologists, and noisy or artifact-contaminated channels were excluded prior to analysis. The dataset offers high-quality, long-duration recordings that reflect real-world clinical conditions, making it suitable for validating data augmentation strategies beyond the CHB-MIT dataset.

A.2. Experimental settings

The experimental protocol used for the SWEC-ETHZ dataset was kept consistent with that of the CHB-MIT dataset to enable a meaningful comparison. Specifically, we applied the same set of 12 data augmentation methods, and each method was used to increase the training data volume by a factor of 100. Feature extraction procedures, classifier configurations, and evaluation metrics remained unchanged.

A.3. Results and consistency analysis

The performance comparison on the SWEC-ETHZ dataset is summarized in Table A.1. The results show that Scaling, ScalingMulti, and MagWarp consistently achieved better classification performance across different classifiers, particularly in terms of Sensitivity and AUC. In contrast, Jittering and Permutation produced limited or negative effects, which is consistent with the findings on the CHB-MIT dataset.

These consistent results across two independent datasets support the robustness and general applicability of our conclusions. The alignment of findings indicates that the effectiveness of data augmentation methods such as Scaling, ScalingMulti, and MagWarp is not restricted to a specific dataset. This additional validation enhances the credibility of the comparative analysis presented in the main study.

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